

- 1) Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?

Ans:- I would use a Bayesian Network to represent relationships between machine parameters like temperature, vibration, and pressure. Using historical data, the model learns the probability of machine failure. When new sensor values come in, we can calculate the probability of failure. If the probability is high, we can perform preventive maintenance. The main advantage is that it can handle uncertainty and combine different sensor information.

- 2) Imagine you are asked in an interview to design an image denoising solution. How would Markov Random Fields help model neighborhood pixel dependencies?

Ans:- I would use a Markov Random Field because neighboring pixels are usually related. Each pixel can be represented as a node, and nearby pixels are connected. The model compares the noisy pixel with its neighbors and finds the most likely clean value. This helps remove noise while preserving important image structures like edges.

- 3) If a company wants to build a speech recognition system, how would you apply a Hidden Markov Model to process sequential audio signals ?

Ans:- An HMM is useful for speech recognition because speech is sequential. The audio features are the observations, while phonemes or words can be treated as hidden states. The model learns how states transition and how likely each observation is for a state. Then an algorithm like Viterbi finds the most likely sequence of states and helps recognize the spoken words.

- 4) During a technical interview, how would you explain the use of Conditional Random Fields for sequence labeling tasks such as part-of-speech tagging ?

Ans:- A Conditional Random Field is used for sequence labeling. In POS tagging, it predicts labels like noun, verb, or adjective for each word. It considers the current word, surrounding words, and neighboring labels. Because it considers context instead of treating each word independently, it can produce more accurate and consistent POS tags.